Hungarian Housing Intelligence

Detailed Project History | November 2026
From a single-city Python web scraper to an interactive real estate analytics platform

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1. The Beginning: The First Real Estate Data Collector | July 2024

The project began in July 2024 as a Python application focused on understanding the real estate market in Debrecen. The original idea was to collect apartment listing data repeatedly and use it to build an index that could be analyzed over time.

One key methodological choice was to focus on apartments measuring 50–60 m², reducing some of the variation that arises when comparing properties of very different sizes. The first implementation transformed unstructured listing pages into structured observations suitable for further analysis.

- Mapped listing-page HTML and sent HTTP requests using Python Requests.
- Extracted property prices and floor areas using BeautifulSoup, converted values into numerical data, and calculated prices per square meter.

2. Building the Historical Database | October 2024

The next objective was persistent time-series storage. The original Debrecen dataset was stored in the MySQL table real_estate_market_data, which recorded observation dates, HUF- and EUR-denominated price-per-square-meter measures, and exchange-rate-related comparison values.

The project expanded beyond web scraping into SQL queries, record insertion, duplicate handling, and maintenance of a growing historical series. This laid the groundwork for comparing housing-price movements over longer periods.

3. Moving Data Collection to the Cloud | March 2025

To reduce the need for manual local execution, the scraper was moved into AWS Lambda, while Amazon DynamoDB provided cloud-based storage. The Python application interacted with AWS using Boto3.

Scheduled execution through Amazon EventBridge allowed the collection process to run periodically without the developer manually launching it on a personal computer. The precise date of this migration is not documented.

4. The First Desktop Analytics Application | November 2025

Alongside the data pipeline, a desktop interface was developed. An initial Tkinter-based application was later given a more modern CustomTkinter interface. The purpose was to make historical observations easy to query, explore, visualize, and hand over to other analytics tools.

The desktop application became the main local control panel for the project, connecting data retrieval and presentation.

- Synchronized cloud data from DynamoDB into local MySQL.
- Exported datasets to Excel and Power BI-compatible formats.
- Created historical charts using Matplotlib, including comparisons between the housing-price index and HUF/EUR exchange-rate measures.

5. Scaling from Debrecen to 47 Locations | March 2026

The system expanded from a single city to 47 geographical locations: all 23 districts of Budapest and another 24 Hungarian cities. The URL generator in multiple_data_fetcher.py produced the appropriate Jófogás query for each target location.

For each location, the scraper processed listing prices per square meter for 50–60 m² apartments. The lower 10% of the observed values were removed, and the median of the remaining observations was used as the location-level price indicator.

This development transformed a Debrecen-focused data collector into the foundation of a platform capable of tracking multiple housing markets in parallel.

- Key output fields: location, date, huf_per_sqm, sample_count, and source_url.
- The expanded output was integrated into the AWS Lambda / DynamoDB collection process.

6. A New Data Model and Synchronization Pipeline | March 2026

The nationwide expansion required a new local MySQL schema: real_estate_47loc, with the 47loc_table table. Records use UUIDv7 identifiers stored in BINARY(16) format. A composite uniqueness constraint on (snapshot_date, location) prevents multiple rows for the same location and observation day.

A separate Python synchronization module retrieves DynamoDB records using paginated scans, normalizes date fields and DynamoDB Decimal values, and then writes records to MySQL in batches using INSERT ... ON DUPLICATE KEY UPDATE.

This separated cloud-based collection from local relational storage, making the local database suitable for analysis, export, and desktop visualizations.

7. Unifying Legacy and Multi-Location Data | July 2026

The older Debrecen history and the new 47-location observations lived in separate MySQL databases. The updated DataGateway component offered a consistent query interface and normalized output into Pandas DataFrames.

For Debrecen, the application uses the original historical series until the first available observation in the new dataset, avoiding overlapping dates. Other locations are read exclusively from the new 47-location table.

The CustomTkinter dashboard introduced city selection, record previews, Excel and Power BI exports, and a view showing the latest observations across locations.

8. More Advanced Analytics and Technical Indicators | September 2026

The visualization layer grew beyond ordinary line charts. Weekly open, high, low, and close (OHLC) observations were calculated from the time series, enabling candlestick-style charts. The analytics module also included 4- and 12-week moving averages, 20-week Bollinger Bands, RSI, MACD, and multi-panel chart layouts.

These candles describe changes in an aggregated asking-price indicator, not individual property transactions or tradable asset prices. The technical indicators are exploratory analysis tools, not reliable stand-alone real estate forecasts.

9. Maintaining the Scraper After Website Changes | October 2026

Changes to Jófogás page markup required a revision of the scraper. Previously used CSS selectors no longer matched the relevant listing data reliably. Saved HTML files and HTTP diagnostics were used to inspect the new structure and identify parsing failures.

The revised extraction logic searched within listing cards, deduplicated listing URLs, validated both price and area values, and explicitly marked unsuccessful or empty extraction results. This stage highlighted a central engineering challenge of web scraping: external data-source structures can change and require ongoing maintenance.

10. Publishing an Interactive Web Dashboard | November 2026

The next milestone was the Hungarian Housing Intelligence interactive web dashboard built with Streamlit and Plotly. The goal was to make the accumulated findings accessible through a browser rather than only through the local desktop application.

The public-facing application reads a prepared CSV snapshot exported from MySQL. It does not require direct public access to the private MySQL database or AWS services. The code was published on GitHub and the dashboard deployed through Streamlit Community Cloud.

- Location selection and date-range filters, current index values, percentage changes, and sample sizes.
- Interactive time-series charts, weekly OHLC candlesticks, and optional technical indicators.
- Location rankings, comparative charts, CSV downloads, and methodology notes.

Current System Architecture

- Data collection: Jófogás → Python scraper → AWS Lambda → DynamoDB
- Storage and processing: DynamoDB → Python synchronization → MySQL → DataGateway → Pandas
- Local analytics: CustomTkinter → Matplotlib → Excel / Power BI
- Web presentation: MySQL → CSV snapshot → Streamlit → Plotly → Streamlit Community Cloud

Results, Limitations, and Future Directions

Over approximately two and a half years, the project brought together web scraping, database design, cloud automation, SQL, desktop development, data visualization, and public web deployment. Rather than ending with a static visualization, it developed into an end-to-end collection and analytics workflow.

Scope and methodology: The roughly 2.5-year historical series belongs to Debrecen; collection across all 47 target locations began later. The index is derived from property asking prices and a custom trimming-and-median procedure. It is not an official transaction-price index, and small samples, listing composition, and source changes can affect its reliability. Some intermediate project dates are approximate.

Future directions include richer cross-city comparisons, stronger data-quality checks and statistical reliability measures, additional visualizations, and automated updates of the published web dataset.

Project Links

Live dashboard: https://hungarian-housing-intelligence.streamlit.app

GitHub: https://github.com/GregTakacsGergo/hungarian-housing-intelligence

The project began in 2024 with a question: how can changes in the housing market be measured using independently collected data? By 2026, it had grown into a working, self-developed data platform — with further development still ahead.